Rosiglitazone in the Assistance of Metabolic Control during Olanzapine Administration in Schizophrenia: A Pilot Double-blind, Placebo-controlled, 12-week Trial

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Abstract

Introduction: Excessive body weight gain (BWG), hyperglycemia and dyslipidemia are important side effects of olanzapine. We assessed the effects of rosiglitazone on BWG, the insulin resistance index (HOMA-IR), lipids, glycated hemoglobin and fibrinogen in olanzapine-treated schizophrenia patients.

Methods: Thirty patients taking olanzapine (10–20 mg daily for 8 months) were randomly allocated to rosiglitazone (n = 15; 4 to 8 mg daily) or placebo (n = 15) in a 12-week double-blind protocol. Anthropometric and biochemical variables were evaluated at baseline, weeks 6 and 12.

Results: The rosiglitazone and placebo groups gained 3.2 ± 4.5 and 2.2 ± 2.3 kg, respectively ($p = 0.65$). Insulin and the HOMA-IR significantly decreased after rosiglitazone ($p < 0.05$). Rosiglitazone did not improve the lipid profile, fibrinogen and Hb1c levels.

Discussion: The positive impact of rosiglitazone was limited to improved glycemic control. It cannot be recommended for metabolic control during olanzapine treatment.

Introduction

There is general agreement that people with schizophrenia or bipolar disorders need more attention to their physical health problems, including appropriate management of the adverse metabolic events associated with psychiatric medications [23]. Olanzapine is an effective agent in schizophrenia and other psychotic disorders [18], but it is linked to clinically significant body weight gain (BWG), hyperglycemia and hyperlipidemia [10, 18, 23]. In the Clinical Antipsychotic Trials of Intervention Effectiveness study (CATIE), a prospective project funded by the US National Institutes of Health comprising 1493 patients with schizophrenia, olanzapine significantly increased body weight (BW), glycated hemoglobin, triglyceride and total cholesterol levels compared to other atypical (quetiapine, risperidone and ziprasidone) and conventional (perphenazine) antipsychotics (APs) [18].

Collectively, olanzapine (and also clozapine) negatively affect the anthropometric and biochemical variables that define the metabolic syndrome [23]. Thus, improved pharmacological and non-pharmacological treatments to assist metabolic control in olanzapine-related patients are urgently needed.

Some clinical trials aimed at preventing or reversing olanzapine-induced BWG have been conducted with nizatidine, amantadine, memantine, topiramate, reboxetine, sibutramine and metformin [4, 8, 27]. Collectively for these pharmacological agents, a quantitative review showed that the figures for BWG prevention and reversal, respectively, in schizophrenia were −1.16 kg, 95% CI [−1.9 to −0.41] and −3.85 kg, 95% CI [−4.25 to −3.44] [8]. By contrast, few randomized clinical trials have been published so far on the specific pharmacological assistance of olanzapine- and other atypical AP-induced hyperglycemia or dyslipidemia [6, 7].

Since the metabolic dysfunction observed during olanzapine administration is linked to insulin resistance [1, 15, 30], a sound strategy has been the use of antidiabetic agents. These compounds might also display neuroprotective effects which could help patients with psychosis [20]. Metformin, which improves glycemic control mainly by decreasing the hepatic glucose output [33], prevented olanzapine-induced BWG in drug-naïve, first-episode schizophrenia patients [34] but not in subjects with a chronic disease in their

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first exposure to the antipsychotic [3]. Besides, metformin decreased BW in schizophrenia patients undergoing prolonged olanzapine administration [4,6]. When considering the carbohydrate and lipid metabolism during olanzapine administration, metformin improved or prevented further impairment in insulin sensitivity level as assessed by the HOMA-IR index [3,4,6,17,34]. By contrast, results in the lipid profile were inconsistent, since metformin decreased total cholesterol and increased HDL levels in one study [3] but not in another [4]. Metformin decreased triglyceride levels in one study [6] but not in other trials [3,4].

The thiazolidinediones (rosiglitazone and pioglitazone) are potent insulin sensitizers widely used in the treatment of type 2 diabetes mellitus. These agents have a different mechanism of action than metformin, since they act by activating the nuclear receptor peroxisome proliferator-activated receptor gamma (PPAR-gamma) in target tissues for insulin [32,33]. Their pharmacodynamic and pharmacokinetic profiles [28] appear to be safe for psychiatric patients who are often in a polypharmacy regimen.

Previous studies have shown that rosiglitazone increases BW in non-psychiatric patients with type 2 diabetes [15,29,35]. Rosiglitazone-induced BWG is associated to body fat accumulation/redistribution and fluid retention [14,29,35]. The same effect is expected during olanzapine administration. However, this has not been reported so far. Besides, the molecular basis of olanzapine-induced insulin resistance has not been clarified [1,15,30]. Explanations of rosiglitazone-induced BWG focus on the fat accumulation linked to improved peripheral insulin sensitivity. As a matter of speculation, it is unknown whether olanzapine also impairs the anorectic effects of insulin in the brain. Were that the case, and perhaps contrarily to the standard diabetic patient, rosiglitazone, might restore weight balance by improving brain insulin receptor sensitivity during olanzapine administration.

In this pilot study the authors aimed to assess the impact of rosiglitazone on body weight (BW), waist circumference, insulin resistance, glycated hemoglobin (Hb1c) and serum lipid levels in patients with schizophrenia after prolonged olanzapine administration. In addition, we assessed fibrinogen levels, since they are positively correlated to the degree of insulin resistance [9].

Patients and Methods

▼

Subjects

This was a 12-week study conducted in inpatients at the Psychiatric Institute "Dr. Raúl Castillo", Peribeca, Táchira State, Venezuela. The ethic committees of this institution and the Venezuelan government (FONACIT) approved the protocol. The study was conducted in a cohort of patients that was switched from conventional APs to olanzapine 8 months before the present investigation. Recruitment was conducted among those subjects willing to lose BW or prevent excessive BWG after an educational program of healthy life style. Voluntary participation was requested and a written informed consent was obtained from each patient and from the medical director of the institution.

The primary outcomes were the change in anthropometric (BW, body mass index and waist circumference) and biochemical variables (fibrinogen, glucose, insulin, Hb1c and lipids and an insulin resistance index (HOMA-IR).

For inclusion into this study, participants must have been older than 18 years, free of hormone replacement and any chronic disease besides the mental disorder and have normal physical and laboratory tests (liver, kidney, thyroid tests) before starting olanzapine treatment and at baseline. Patients were under olanzapine monotherapy, but benzodiazepine use was allowed to improve sleep and to lessen agitation. Recommendations for healthy food and physical exercise to control BWG were provided at the beginning of the study. It was not possible to quantify appetite and daily food consumption.

Thirty patients with schizophrenia entered the study. The psychiatric diagnosis was conducted by research psychiatrists or social workers using the Structured Clinical Interview for the DSM-IV [31]. The BWG (kg, mean) under olanzapine before the trial was: 4.2; SD: 1.9; median: 4.0; range: 0.6–7.5 kg (1.7–12.7% of baseline). Treatment duration with olanzapine before entering the study was 8 months for all subjects. The daily olanzapine dose (mg) was: mean: 10.51; SD: 2.04 mg; median: 10; range: 10–20. No differences in previous BWG ($p = 0.33$) or olanzapine dose ($p = 0.19$) were observed in the subjects assigned to the rosiglitazone and placebo groups. Data on the family history of diabetes and cardiovascular disease was not considered as reliable. Therefore, it was not included in the final analysis.

Subjects were randomly assigned through a computer-based program to either rosiglitazone (Avandia, Glaxo-Smith-Kline, Caracas, Venezuela) or placebo.

Procedure

Medication and placebo were administered as a single daily dose by nurses trained in research and the number of pills was weekly quantified. Rosiglitazone was started at the lower dose (4 mg), and at week 4 the full rosiglitazone (8 mg) or placebo dose was achieved and maintained thereafter in all subjects. The rosiglitazone schedule encompassed the usual dose range in clinical practice. Since rosiglitazone was devoid of significant side effects, no dose adjustment was necessary.

The mental status was monitored with the Brief Psychiatric Rating Scale (BPRS) at baseline, at week 6 and at the end of the study.

Assessments

Anthropometric variables

Each subject's blood pressure, BW and body mass index (BMI = weight in kg/height squared) were measured at baseline, weeks 6 and 12. The waist circumference (1 cm above navel) was assessed at baseline and at week 12. Edema quantification was not carried out. All measurements were conducted in strict fasting conditions.

Biochemical variables

Cubital vein blood samples were drawn in fasting conditions at baseline, weeks 6 and 12 to assess glucose and lipids. Fibrinogen, insulin and Hb1c were only assessed at baseline and at week 12. An index of insulin resistance (HOMA-IR) was calculated as follows: glucose (mmol/L) × insulin mIU/mL)/22.5 [19]. Liver function (serum levels of transaminases, free and conjugated bilirrubin and lactic dehydrogenase (LDH) was only assessed at the end of the study.

Chemical analysis

Serum glucose, lipids, transaminases, bilirrubin and LDH were quantified with an enzymatic method from Boehringer (Germany). Insulin was assessed in duplicate by radioimmunoassay with commercial kits from DPC, Los Angeles, CA, USA.

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The inter- and intra-assay variation was <10%. Fibrinogen was assessed with the thrombin-clotting method from Sigma (St. Louis, Missouri, USA) (detection limits 50 mg/dL). Hb1C levels were measured by a reflectometry method from Nyco Card (Cyprus; reference values: 4.5–6.3%).

Statistical analysis

The SPSS 13.0 version (SPSS Inc. Chicago, Ilinois, USA) was used. Data normality was estimated with the one-sample Kolmogorov-Smirnov test and Levene's test for the equality of variances. For waist circumference, insulin, the HOMA-IR, fibrinogen and Hb1c levels, within and between group comparisons were conducted with the two-tailed t-test for related and unrelated samples, respectively. Data on body weight, BMI, glucose and lipids were analyzed with the General Lineal Model, with time and treatment as within and between factors, respectively. Bivariate correlation analysis was conducted with the Pearson and Spearman coefficients. Frequencies were analyzed with the chi-squared test. Results were considered significant at the $p < 0.05$ level.

Results

Demographic features

One subject in the rosiglitazone group left the study in the first week of treatment due to a change in residence. Hence, it was not possible to conduct either an "intention to treat" or a "last observation carried forward" analysis. All the placebo-treated subjects completed the study. ▶ **Table 1** describes the demographic and clinical features of the completers.

All subjects engaged in healthy food and monitored physical exercise programs but their effects could not be quantified. Rosiglitazone was well tolerated at the maximal dose in all patients.

Table 1 Gender and age in the treatment groups.

Treatment	Gender	Age (years)
rosiglitazone (n = 14)	males: 8 (57.1%)	40.9 ± 9.2
	females: 6 (42.9%)	48.7 ± 9.2
placebo (n = 15)	males: 6 (40%)	55.3 ± 6.9
	females: 9 (60%)	48.8 ± 6.7

The gender distribution was similar in both groups: χ^2 (1) = 0.8, p = 0.5. Men in the placebo group were significantly older than men in the rosiglitazone group: f (3, 28) = 3.8, p = 0.023

No specific complaint could be attributed to the drug treatment. The BPRS significantly decreased over time (p = 0.002) but the figure did not differ between the groups (p = 0.29) (▶ **Table 2**). Blood pressure (mmHg) did not change significantly in any treatment group, (p = 0.8 for the between group comparisons, data not shown).

Anthropometric variables

Body weight and the BMI significantly increased after both treatments without significant differences between the groups (p = 0.65 and 0.47, respectively) (▶ **Table 3**). The waist circumference increased in both groups but only reached significance after placebo (p = 0.046).

Biochemical variables

Carbohydrate metabolism and fibrinogen

Glucose level changes did not differ significantly between the groups (p = 0.7, ▶ **Table 2**). Insulin and the HOMA-IR index significantly decreased after rosiglitazone administration (p = 0.005 and 0.004, respectively, for the within group comparisons, and 0.037 and 0.027, respectively, for the between group comparisons). Hb1c levels increased in both groups but only reached significance after placebo (p = 0.02) with no differences between the groups (p = 0.8). Fibrinogen levels increased in both groups but only reached significance after rosiglitazone (p = 0.03) with no differences between the groups (p = 0.8) (▶ **Table 3**).

Blood lipids

Total and LDL cholesterol levels displayed a significant decrease over time with both treatments with no between group differences. Considering total cholesterol, such a decrease tended to be smaller after rosiglitazone (p = 0.08, ▶ **Table 4**). The HDL levels remained stable in the placebo group and displayed a non-significant decrease after rosiglitazone (p = 0.27, ▶ **Table 4**). Conversely, triglyceride levels remained stable after rosiglitazone and tended to decrease after placebo. However, the figure did not reach statistical significance (p = 0.4, ▶ **Table 4**).

Effects of gender, basal body mass index and age on body weight

Body weight gain did not differ between the genders in both treatment groups even though it tended to be higher in men after rosiglitazone: 4.3 ± 5.1 (men) vs. 1.5 ± 2.7 (women), p = 0.25

Table 2 Change in BPRS scores, body weight, waist circumference and body mass index after rosiglitazone or placebo.

	Treatment	Basal	Week 6	Week 12	Δ at week 6	Δ at week 12
BPRS[a]	rosiglitazone	21.4 ± 9.6	20.4 ± 8.5	18.0 ± 6.1	−1.2 ± 6.4	−3.4 ± 7.9
	placebo	24.9 ± 7.7	24.6 ± 10.3	18.5 ± 8.6	−0.33 ± 5.9	−6.5 ± 5.7
body weight (kg)[b]	rosiglitazone	63.9 ± 11.1	66.2 ± 11.4	66.9 ± 11.1	2.4 ± 4.5	3.2 ± 4.5
	placebo	61.1 ± 9.7	62.4 ± 9.9	63.3 ± 10.0	1.3 ± 1.5	2.2 ± 2.3
body mass Index (kg/m^2)[c]	rosiglitazone	25.4 ± 4.8	26.6 ± 4.3	26.9 ± 4.2	1.2 ± 2.0	1.5 ± 2.0
	placebo	25.3 ± 2.7	25.9 ± 2.9	26.3 ± 2.9	0.54 ± 0.6	0.92 ± 0.96
waist circumference (cm)[d]	rosiglitazone	89.9 ± 12.5	–	91.5 ± 11.6	–	1.6 ± 3.7
	placebo	88.5 ± 9.1	–	90.5 ± 8.6	–	2.0 ± 3.5

Values represent mean ± standard deviation

[a]Time: f (2, 26) = 7.8, p = 0.002; time × treatment: f (2, 26) = 1.3, p = 0.29

[b]Time: f = 9.9, p = 0.001; time × treatment: f = 0.43, p = 0.65

[c]Time: f = 10.5, p = 0.000; time × treatment: f = 0.8, p = 0.47

[d]Between groups: t (26) = 0.4, p = 0.6; within groups: rosiglitazone: t (13) = 1.6, p = 0.12; placebo: t (14) 2.2, p = 0.046

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Table 3 Change in glucose, insulin, the HOMA-IR index, glycated hemoglobin and fibrinogen after rosiglitazone or placebo.

	Treatment	Basal	Week 6	Week 12	Δ at week 6	Δ at week 12
glucose (mg/dL)[a]	rosiglitazone	85.4 ± 14.0	83.1 ± 13.1	80.6 ± 13.8	−2.3 ± 10.4	−4.8 ± 8.3
	placebo	82.4 ± 9.1	84.5 ± 8.8	78.8 ± 8.7	2.1 ± 9.8	−3.6 ± 5.2
insulin (mIU/mL)[b]	rosiglitazone	24.4 ± 18.9	–	8.6 ± 4.2	–	−15.8 ± 17.6
	placebo	14.7 ± 10.0	–	12.3 ± 13.2	–	−2.4 ± 15.4
HOMA-IR[c]	rosiglitazone	5.1 ± 3.9	–	1.7 ± 0.9	–	−3.3 ± 3.6
	placebo	2.9 ± 1.6	–	2.4 ± 2.8	–	−0.5 ± 2.9
Hb1c (%)[d]	rosiglitazone	5.0 ± 0.8	–	5.5 ± 1.1	–	0.5 ± 1.2
	placebo	4.7 ± 0.7	–	5.4 ± 0.7	–	0.6 ± 1.0
fibrinogen (mg/dL[e]	rosiglitazone	266.4 ± 78.6	–	334.9 ± 117.4	–	68.6 ± 100.6
	placebo	291.1 ± 98.7	–	344.2 ± 100.7	–	53.2 ± 150.8

Values represent mean ± standard deviation
[a]Time: f (2, 26) = 6.15, p = 0.007; time × treatment: f (2, 26) = 0.4, p = 0.7
[b]Between groups: t (27) = 2.2, p = 0.037; within groups: rosiglitazone: t (13) = 3.4, p = 0.005; placebo: t (14) = 0.6, p = 0.6
[c]Between groups: t (27) = 2.3, p = 0.027; within groups: rosiglitazone: t (13) = 3.5, p = 0.004; placebo: t (14) = 0.6, p = 0.5
[d]Between groups: t (27) = 0.2, p = 0.8; within groups: rosiglitazone: t (13) = 1.6, p = 0.12; placebo: t (14) = 2.4, p = 0.02
[e]Between groups: t (27) = 0.3, p = 0.8; within groups: rosiglitazone: t (13) = 2.5, p = 0.03; placebo: t (14) = 1.4, p = 0.1

Table 4 Change in blood lipids after rosiglitazone or placebo.

	Treatment	Basal	Week 6	Week 12	Δ at week 6	Δ at week 12
total cholesterol (mg/dL)[a]	rosiglitazone	192.4 ± 33.8	184.9 ± 32.5	179.9 ± 22.6	−7.5 ± 29.2	−12.6 ± 29.5
	placebo	228.9 ± 39.9	215.5 ± 41.1	201.2 ± 26.02	−13.4 ± 22.9	−27.6 ± 19.7
LDL cholesterol (mg/dL)[b]	rosiglitazone	115.3 ± 24.4	116.9 ± 29.5	108.5 ± 19.9	1.7 ± 19.7	−6.7 ± 20.8
	placebo	149.3 ± 31.7	141.6 ± 35.9	124.5 ± 20.9	−7.8 ± 25.6	−24.9 ± 21.2
HDL cholesterol (mg/dL)[c]	rosiglitazone	44.4 ± 10.0	41.9 ± 10.5	38.7 ± 11.1	−2.6 ± 7.5	−5.7 ± 10.3
	placebo	43.4 ± 15.4	46.5 ± 8.9	44.4 ± 12.8	3.2 ± 11.9	0.98 ± 11.6
triglycerides (mg/dL)[d]	rosiglitazone	163.7 ± 77.3	130.4 ± 43.2	163.0 ± 52.7	−33.3 ± 58.1	−0.7 ± 48.9
	placebo	173.7 ± 91.7	140.1 ± 55.9	161.5 ± 76.7	−33.6 ± 52.2	−12.2 ± 33.4

Values represent mean ± standard deviation
[a]Time: f (2, 26) = 10.2, p = 0.001; time × treatment: f (2, 26) = 1.1, p = 0.08
[b]Time: f (2, 26) = 9.5, p = 0.001; time × treatment: f (2, 26) = 2.0, p = 0.15
[c]Time: f (2, 26) = 2.3, p = 0.12; time × treatment: f (2, 26) = 1.3, p = 0.27
[d]Time: f (2, 26) = 5.6, p = 0.01; time × treatment: f (2, 26) = 0.8, p = 0.4

(▶ **Table 5**). Besides, the BW change did not correlate significantly with the initial BMI: rosiglitazone: r (14) = −0.3, p = 0.24; placebo: r (15) = 0.075, p = 0.79.
A significant negative correlation between BW change and age was observed in the placebo group: r (15) = −0.57, p = 0.027. In the rosiglitazone group, the figure was only a trend: r (14) = −0.43, p = 0.12.

Table 5 Change in body weight (kg) according to gender.

Treatment	Masculine	Feminine	t (p)
rosiglitazone	4.3 ± 5.1	1.5 ± 2.7	1.2 (0.25)
placebo	2.0 ± 2.9	2.3 ± 1.9	0.3 (0.8)

Values represent mean ± standard deviation. *t* (p) = statistics for the *t* test for unrelated samples and its associated probability in parenthesis

Correlation analysis

For each treatment (rosiglitazone or placebo), a correlation analysis was conducted between the BW change and the biochemical variables and BPRS scores at baseline and with the change in these variables after treatment. No figure reached statistical significance (data not shown).

Liver function

Since isolated cases of liver toxicity have been reported during rosiglitazone administration [16], some markers of hepatic function were assessed. Serum levels of transaminases (alanine and aspartate), lactic dehydrogenase, total, unconjugated and conjugated bilirrubin did not differ between both groups at the end of treatment (▶ **Table 6**).

Discussion

The antidiabetic rosiglitazone appeared to be a promising agent as a adjunctive treatment during clozapine or olanzapine administration, since it is highly effective in glycemia control [33] and may have positive effects on blood pressure, kidney function and endothelial protection [25,26]. This last contention was recently challenged by a study suggesting an increase in myocardial infarction and death from cardiovascular causes after prolonged rosiglitazone administration [24].
As recommended in recent literature reviews, all patients received nutritional advice and entered a program of healthy life style [5,8,21]. Rosiglitazone was well tolerated at the maximal dose and did not negatively impact the liver function. However, it did not prevent or reverse BWG. In fact, the placebo group gained less BW, even though the figure did not reach statistical

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Table 6 Markers of liver function after rosiglitazone or placebo.

Treatment	Alanine Transaminase	Aspartate Transaminase	Lactic Dehydrogenase	Total Bilirrubin	Unconjugated Bilirrubin	Conjugated Bilirrubin
rosiglitazone	13.8 ± 14.4	21.4 ± 14.7	127.0 ± 48.5	1.1 ± 0.4	0.71 ± 0.26	0.39 ± 0.19
placebo	13.9 ± 6.9	20.3 ± 7.4	102.4 ± 25.8	1.2 ± 0.5	0.62 ± 0.25	0.53 ± 0.3
t (p)	0.035 (0.97)	0.26 (0.79)	1.7 (0.097)	0.27 (0.78)	0.98 (0.33)	1.4 (0.15)

Values represent mean ± standard deviation. (t) = statistics of the two-tailed *t* test for unrelated samples and its associated probability (p) for 27 degrees of freedom

significance (3.2 ± 4.5 vs. 2.2 ± 2.3 kg, p = 0.65). The waist circumference increased in both treatment groups as well, even though the figure only reached significance in the placebo group (p = 0.046).

Studies in diabetic patients often report BWG after rosiglitazone administration which has been attributed to body fat accumulation and redistribution and fluid retention [14, 29, 35]. This could not be assessed here since we did not measure body composition and the magnitude of edema.

Gender did not significantly affect BWG Interestingly, BWG during rosiglitazone tended to be higher in men (4.3 vs. 1.5 kg), but the relevance of this finding cannot be inferred from our study given the small sample size. We did not detect any significant influence of the initial BMI on BW change during rosiglitazopne or placebo. Unfortunately most subjects had a BMI < 30 (78.6 % and 86.6 % in the rosiglitazone and placebo group, respectively). Hence, a definitive conclusion cannot be stated on this clinically important relationship. The magnitude of BWG correlated inversely with age as it has been shown in other studies [10].

The failure of rosiglitazone in preventing BWG is consistent with numerous studies in diabetic populations [14, 29, 35]. Since the HOMA-IR significantly decreased after rosiglitazone, it is an indirect support of the contention that insulin resistance and hyperinsulinemia are not directly involved in olanzapine-induced BWG. Instead, it has been argued that appetite stimulation is a more proximal mechanism in atypical antipsychotic-induced BWG than any other metabolic drug effect [2].

As expected, the effect of rosiglitazone on carbohydrate metabolism was clearly favourable, since a significant decrease in insulin levels and the HOMA-IR index was observed. However, a more exhaustive analysis of glucose metabolism should have included either an oral glucose tolerance test or a euglycemic-hyperinsulinemic clamp. Importantly, the Hb1c levels increased in both groups, although the figure only reached significance in the placebo group (p = 0.02). Fibrinogen increment after olanzapine administration was not prevented by rosiglitazone either. The same lack of effect on fibrinogen levels was recently reported with metformin in psychiatric patients [4] and non-psychiatric subjects with glucose intolerance [13].

The lipid profile was not favourably impacted by rosiglitazone. While total cholesterol and LDL levels decreased over time in both groups, such a decrease tended to be more pronounced after placebo. The HDL levels numerically decreased after rosiglitazone, while they remained stable after placebo; however, the figure did not reach statistical significance. Finally, triglyceride levels remained stable after rosiglitazone, while they tended to decrease after placebo. Again, the figure did not reach statistical significance. Studies conducted in subjects with type 2 diabetes also showed that rosiglitazone tended to increase triglyceride levels and that the increase in HDL and the decrease in LDL were smaller compared to pioglitazone, a related compound [11, 12].

Relevant limitations of this study were the small sample size, the lack of edema assessment and that factors involved in inflammation and coagulation others than fibrinogen were not assessed.

In conclusion, rosiglitazone improved the insulin resistance index (HOMA-IR), it did not display a positive effect on lipids, fibrinogen and Hb1c and did not reverse or prevent additional BWG during olanzapine administration. Hence, except for its positive impact on insulin resistance, these results do not support the use of rosiglitazone for metabolic control in this clinical population. In addition, recent concerns regarding its safety [24] should be taken into account when evaluating its use in psychiatric patients with multiple risk factors for cardiovascular diseases [22]. Instead, growing evidence supports the use of topiramate, metformin, sibutramine or reboxetine in the assistance of metabolic control during olanzapine administration [2, 8].

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